

Answers to mandatory 1

MAT-MEK4270

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1.2)

3)

We first calculate all second derivatives,

$$\begin{aligned}\frac{\partial^2}{\partial t^2} u &= (-i)^2 \omega^2 e^{i(k_x x + k_y y - \omega t)} = -\omega^2 e^{i(k_x x + k_y y - \omega t)} \\ \frac{\partial^2}{\partial x^2} u &= i^2 k_x^2 e^{i(k_x x + k_y y - \omega t)} = -k_x^2 e^{i(k_x x + k_y y - \omega t)} \\ \frac{\partial^2}{\partial y^2} u &= i^2 k_y^2 e^{i(k_x x + k_y y - \omega t)} = -k_y^2 e^{i(k_x x + k_y y - \omega t)}.\end{aligned}$$

Putting them together, we get

$$u_{tt} - c^2 \Delta u = \underbrace{(-\omega^2 + c^2 \overbrace{(k_x^2 + k_y^2)}^{\omega^2})}_0 e^{i(k_x x + k_y y - \omega t)} = 0,$$

which means that $u = e^{i(k_x x + k_y y - \omega t)}$ satisfies the wave equation.

4)

To make everything fit we immediately start by dividing everything by $e^{i(kh(i+j) - \tilde{\omega} n \Delta t)}$, then take note that in this setup, $\omega = \sqrt{2}kc$, and we calculate

$$\begin{aligned}\frac{e^{-i\tilde{\omega}\Delta t} - 2 + e^{i\tilde{\omega}\Delta t}}{\Delta t^2} &= 2c^2 \frac{e^{ikh} - 2 + e^{-ikh}}{h^2} \\ \cos(\tilde{\omega}\Delta t) &= \frac{2c^2 \Delta t^2}{h^2} (\cos(kh) - 1) + 1 = 2C^2 (\cos(kh) - 1) + 1 = \cos(kh) = \cos\left(\frac{\omega h}{\sqrt{2}c}\right) \\ \Rightarrow \tilde{\omega} &= \frac{1}{\sqrt{2}} \frac{h}{c\Delta t} \omega = \omega.\end{aligned}$$